



POSCO Holdings: Building RAG-based corporate generative AI service using Gemini 1.5 Pro

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- ✓ [Gemini 1.5 Pro](#) model delivers high LLM performance and efficiency
- ✓ Possibility to build various enterprise AI services with flexibility in over 100,000 news reports
- ✓ Keeps employees up to date by providing answers with up to 95% accuracy rate

POSCO Holdings' AI service utilizes Google Cloud's data solution and Gemini 1.5 Pro to help organizations analyze market trends and answer questions.

[POSCO Holdings](#) is leading innovation through new technologies with its Future Technology Research Institute using artificial intelligence (AI), and is planning scenarios to identify and effectively introduce cutting-edge technologies, including robots, as seen by the industry. With AI emerging as an important keyword, the company devised an AI strategy through the AI Research Institute in 2022 and is innovating material development and manufacturing processes through a model that fits its business.

The company recently developed a reporting system and knowledge search portal that allows users to read technology and industry trends at a glance with the help of generative AI (gen AI), using [Google Cloud](#) and [Gemini 1.5 Pro](#). Since 2022, the Institute has improved the production process of long-standing manufacturing industries such as steel, and used batteries, and created processes appropriate for newer industries as well. The institute is also contributing to the spread of gen AI and related advanced AI technologies developed by building and expanding the system for other applications within the group.



Creating LLM-based services that deliver valuable insights and knowledge

The Applied AI Research team at the Future Technology Research Institute aims to enhance its members' capabilities through effective information acquisition and knowledge asset management. In a fast-changing market, understanding the latest global trends in the materials field is very important for research, development and production, and strategy formulation. Until now, the common practice is for employees to gather information from news sources and websites to identify trends, and collaboratively share, review, and present this information to one another.

To improve the process, POSCO Holdings now has a “Material News Trend Reporting” model that automatically collects material-related information based on key information and sources from around the world. Using gen AI, the information is then translated as needed, and delivered to employees’ emails on a regular basis as a report that includes recommendations, text summaries, and country-specific trends.

The Applied AI Research team has even gone one step further by adding a service that continuously collects and learns related news and expert knowledge and processes it into secondary knowledge information they call the Knowledge Search and Q&A System. Just like using a general conversational language model, the knowledge search model provides accurate information retrieval when users enter personalized material-related information and interests in natural language, and produces immediate answers with interpreted insights.

Through these two AI-based services, POSCO Group executives and employees can understand market trends in real time without having to manually search for the latest news, and can stay informed with the latest market trends quickly and accurately. Additionally, the more than 100,000 pieces of carefully selected news and reports in the repository will not disappear from the platform, but instead be stored as knowledge assets that can be used at any time through the Knowledge Search and Q&A System.



Building a high-performance, advanced RAG and Gemini-based AI assistant

While gen AI produces information quickly and in large volumes, it faces known security issues and must also address hallucination problems to deliver the high overall performance needed for practical applications in critical business areas.

POSCO Holdings' knowledge search and Q&A system boasts an average accuracy of over 91% in various application domains. It can be viewed as a reliable source of knowledge needed for work, and due to the inherent nature of AI, its accuracy continues to improve consistently.

Specifically, to build a high-performance knowledge Search and Q&A System, the Applied AI Research team

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combined document integrated processing AI as well as advanced search augmentation generation (Advanced RAG) by combining the Multi-stage AI Agent concept and Multi-view Multi-scale Retrieval method. The RAG model was developed in-house to achieve high accuracy. By combining the high-performance AI architecture developed in this way with Google Cloud's Gemini 1.5 Pro, we have built a unique knowledge AI agent system. The material trend knowledge search and Q&A system built through this provides accurate knowledge searches and answers with a high accuracy of 95% even in a workbook database of hundreds of thousands of pages containing over 100,000 recommended news and carefully selected reports.

than 30,000 pages related to electrical equipment. As with previous projects, we integrated our advanced RAG technology with the Gemini 1.5 Pro model, resulting in a system that delivers significantly better performance.”

Woosang Lim

Research Fellow and
Executive Director,
POSCO Holdings Future
Technology Research
Institute

Gemini 1.5 Pro showcased exceptional performance and efficiency when integrated with POSCO Holdings' high-performance search-augmented creation architecture. Specifically, POSCO Holdings effectively managed data flows by developing this architecture alongside the data pipeline of its in-house Applied AI platform, leveraging Google Cloud's advanced data and AI solutions, including [Data Catalog](#), [BigQuery](#), [Vertex AI](#), and [Apigee](#). In conjunction with Gemini 1.5 Pro, this collaboration resulted in the creation of an optimal environment for processing knowledge information data.

The unique knowledge AI agent system developed serves as a solid foundation for easy and effective expansion to other applications. Recently, POSCO Holdings collaborated with POSCO to create a high-performance version of the “Facility Electrical Knowledge Q&A”, achieving an accuracy rate of more than 91%. “We

implemented multi-modal AI capable of seamlessly interpreting POSCO's diverse unstructured documents, including tables and images, to process more than 30,000 pages related to electrical equipment. As with previous projects, we integrated our advanced RAG technology with the Gemini 1.5 Pro model, resulting in a system that delivers significantly better performance.”



Building an efficient generative AI application system using Google Cloud

In building a RAG-based gen AI service that handles complex and enormous data, POSCO Holdings is able to bring its ideas to reality with Google Cloud AI solutions based on effective data management, application of various models, and strong infrastructure performance. The company is continuously anticipating new possibilities while solidifying its AI technological foundation with Google Cloud.

In addition to gen AI-based knowledge search and Q&A systems, POSCO Holdings is continuing research on multimodal gen AI and AI-based simulation to accelerate materials development. The purpose is to create a research environment through AI that can examine various possibilities by testing the properties and changes of various materials in advance and produce new materials more quickly with precision. POSCO Holdings has already demonstrated its technical expertise in achieving highly stable results in the creation of crystal material structures using large-scale language models.

“POSCO Group has continued to research and invest in AI applications over the past few years, and has recently also worked on developing and applying gen AI application algorithms. In order to properly utilize our own AI model, we needed an infrastructure that could effectively serve data and models. Google Cloud helped us quickly and efficiently lay the foundation for gen AI innovation through its outstanding AI capabilities, abundant experience, and close technical support from engineers,” said Lim.

POSCO Holdings also plans to build an AI agent system and an autonomous laboratory environment to make it easier to use Google Cloud AI services. Lim says, “As gen AI technologies advance rapidly and exhibit significant potential, it is crucial to make efficient

efforts to develop high-performance gen AI systems for effective business application. By leveraging Google Cloud to provide the necessary services and infrastructure, along with the flexible application of new technologies, we can uncover greater possibilities.”

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Woosang Lim

Research Fellow and Executive Director, POSCO Holdings Future Technology Research Institute

POSCO was founded as a steelmaking business that extended into other sectors, serving as the backbone of economic and social development in Korea. Recently, they have ventured into rechargeable battery materials, helping to place a renewed global identity of POSCO Group.

Industry: Technology

Location: South Korea

Products: [Gemini 1.5 Pro](#), [Vertex AI](#), [Apigee](#), [BigQuery](#), [Data Catalog](#)

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